

aRboretum : Modernizing botanical labels : towards an inclusive and automated scientific mediation

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Abstract

As biodiversity continues to decline at an alarming rate, raising public awareness has become a crucial seed from which individual consciousness can grow and collective action can flourish. This internship report presents aRboretum, an open-source R package designed to transform botanical garden labels into effective scientific outreach tools. In response to the limitations of traditional plant labels and the urgent need for new approaches to public engagement, aRboretum automates the production, digitization, and enrichment of multilingual botanical labels. It integrates reliable taxonomic data, generates descriptions in multiple languages, and incorporates both audio and visual features. Designed as a modular tool, aRboretum streamlines label production, improves accessibility, and promotes inclusion by providing a participatory scientific communication platform adaptable to botanical gardens, education, biodiversity conservation, and many other applications.

Keywords

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References

Mounce, Ross, Paul Smith, and Samuel Brockington. 2017. 'Ex Situ Conservation of Plant Diversity in the World's Botanic Gardens'. *Nature Plants* 3 (October). <https://doi.org/10.1038/s41477-017-0019-3>.

1. Introduction

Botanical gardens, such as the Jardim Botânico do Rio de Janeiro (JBRJ), founded in 1808 and a UNESCO World Heritage Site (UNESCO Heritage Centre), play an essential multifunctional role for ex situ conservation, environmental education and taxonomic research. According to the International Union for Conservation of Nature (IUCN), these institutions maintain and house more than 100,000 plant species, or about 30% of the world's vascular flora with an overrepresentation of endangered species (Mounce et al. 2017). The JBRJ participates in this effort by sheltering more than 8,000 species, many of which are endemic to Brazil (JBRJ), making it a vital reserve of native flora for unique ecosystems such as the Mata Atlântica, the Cerrado or the Amazon.

2. Methods and Results

3. Conclusion